

Preferential flow and pesticide transport in a clay-rich till: Field, laboratory, and modeling analysis

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[1] This study investigates vertical flow and pesticide transport along fractures in water saturated unoxidized clayey till. From two experimental fields, each 40 m², 96% and 98%, respectively, of total vertical flow was conducted along fractures in the till, while the remaining 2–4% of flow occurred in the clay matrix at very slow flow rate. An applied dye tracer was observed only along 10–26% of the total fracture length measured on the horizontal surface of the experimental fields. In vertical sections the dyed fracture portions constituted root channels, which penetrated the till vertically along the fractures into the local aquifer at 5 m depth. No dye tracer was observed in the fractures without root channels or in the unfractured clay matrix, suggesting that root growth along the fracture surfaces was the principal agent of fracture aperture enhancement. Using hydraulic fracture aperture values determined from large undisturbed column (LUC) collected from one of the experimental fields, it was estimated that 94% of flow in the fractures was conducted along the fracture root channels, while only 6% of flow was conducted along the fracture sections without root channels. For natural vertical hydraulic gradients (0.8–2.3 at the site), flow rates of 0.8–2 km/d were determined for a fracture root channel, while fracture sections without root channels revealed flow rates of 9–22 m/d. Corresponding flow rates in the unfractured matrix were 7–19 mm/yr. For infiltrated bromide (nonreactive tracer) and mobile pesticides mecoprop (MCPP) and metsulfuron, very rapid migration (0.28–0.5 m/d) and high relative breakthrough concentrations (30–60%) into the aquifer were observed to occur along the fracture root channels using a constant hydraulic gradient of 1. Only traces were measured from infiltration of the strongly sorbed pesticide prochloraz. The concentrations of the bromide and pesticides in the monitoring wells were modeled with a discrete fracture matrix diffusion (DFDM) model coupled with a single porosity model (SP) for the till and aquifer, respectively. Using effective fracture spacings and mean fracture apertures for the fracture channel sections as modeling input parameters for the till, the concentrations observed in the wells of the aquifer could be reasonably approximated.

INDEX TERMS: 1803 Hydrology: Anthropogenic effects; 1829 Hydrology: Groundwater hydrology; 1831 Hydrology: Groundwater quality; 1832 Hydrology: Groundwater transport; **KEYWORDS:** clay, fractures, root channels, preferential flow, contaminant transport, dye tracer

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1. Introduction

[2] Glacial deposits of clay-rich till constitute the landscape and the upper geological layers in large parts of the

Northern Hemisphere's agricultural and rural areas. Groundwater used for drinking water in these areas is often extracted from aquifers, interlayered or overridden by the tills. Over the last decade intensive monitoring of such aquifers in Denmark, and elsewhere, has revealed an increasing number of water extraction wells contaminated by, e.g., pesticides and chlorinated solvents from various diffuse and point contaminant sources [Pankow and Cherry, 1996; GEUS, 1999]. The finding of the contaminants in the groundwater may be the result of rapid preferential flow in the overriding

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aquitards and their topsoils. Preferential transport in the aquitards has been observed to occur as the result of (1) abundant conductive biopores (formed by plant roots and small soil burrowing animals) in the soil zone, (2) hydraulically conductive fractures to at least 10 m depth in the subsoil till deposits, (3) vertically connected thin sand layers, (4) seasonally high groundwater tables allowing preferential flow processes to control contaminant transport from shallow depth in the geological profile (5) small contributions to flow from the unfractured soil because of very low hydraulic conductivities typically found for the clayey till material [Jørgensen and Fredericia, 1992; McKay *et al.*, 1993a; Traub-Eberhart *et al.*, 1995; Brown *et al.*, 1995; Jørgensen *et al.*, 1998a, 1998b; Broholm *et al.*, 1999a, 1999b; Jensen *et al.*, 1998a, 1998b, 1999].

[3] Shallow fractures in the tills are formed by desiccation while deeper fractures are mainly formed by stress forces due to glacial ice loadings or tectonic movements in the bedrock [Jørgensen and Fredericia, 1992; McKay and Fredericia, 1995; Klint and Gravesen, 1999; Jacobsen and Klint, 1999]. Investigations using large undisturbed columns (LUC) collected in vertical profiles from 1 to 6 m depth from 8 till sites in Denmark shows a tendency for fracture apertures to decrease with depth [Jørgensen *et al.*, 1998b]. A similar tendency has been reported for a Canadian lacustrine clay by O'Hara *et al.* [2000], and fits the conceptual model that fracture apertures are likely to decrease with depth because of increase in lateral stress from overburden load. Superimposed on this tendency, the Danish column studies reveals that root channels occasionally form highly conductive flow paths to at least 6 m depth along otherwise very narrow or closed fractures [Jørgensen *et al.*, 1998b]. These observations suggest that large aperture fractures occupied by root channels may have greater influence on water flow to greater depth than normally assumed.

[4] Numerical groundwater flow and transport models are needed to assess the risks related to downward migration of pesticides as well as numerous other contaminants in the fractured tills. In previous studies discrete fracture flow/matrix diffusion (DFMD) models have provided reasonable descriptions of the rapid vertical flow and solute tracer transport that has been observed along fractures in lab columns of the tills [e.g., Jørgensen *et al.*, 1998a; Broholm *et al.*, 1999a, 1999b]. In the DFMD models the fractures and matrix domains representing the mobile and immobile water, respectively, are explicitly represented and may be parameterized directly from geological observations. However, obtaining appropriate fracture and matrix characteristics from the clay tills for use in DFMD models has been a major challenge. Progress has recently been made from extensive mapping of fractures in tills [Klint and Gravesen, 1999; Jacobsen and Klint, 1999] and laboratory studies using LUC taken from Danish and Canadian clays [Jørgensen and Spliid, 1993; Hinsby *et al.*, 1996; Hoff, 1998; Jørgensen *et al.*, 1998a, 1998b, 1998c; Jørgensen and Spliid, 1998; Jensen *et al.*, 1998a, 1998b, 1999; Helstrup, 1998; Thorsen *et al.*, 1998; Broholm *et al.*, 1999a, 1999b; Jørgensen *et al.*, 2000; O'Hara *et al.*, 2000]. In one experiment by Jørgensen *et al.* [1998a], the physical characteristics of the till from the current study site were measured or estimated from LUC, including hydraulic conductivity, fracture spacing, fracture aperture (determined by the cubic law),

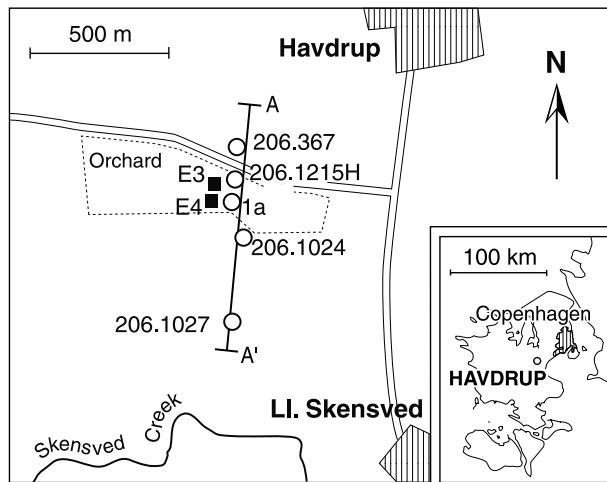
and matrix porosity. Tracer experiments were then carried out using nonreactive solute (chloride) and the pesticides mecoprop (MCP) and simazine. Transport was simulated using both single porosity (SP) and DFMD approaches. The simulations showed that the DFMD model, calibrated to the flow data, was capable of accurately predicting transport of both chloride and the pesticide MCP at the column (LUC) scale, while the SP model underestimated the transport rates with more than 1 order of magnitude. Although this study was promising, further investigation was needed to determine whether DFMD models calibrated to LUC transport data could be validated at the field scale in order to use such calibrated models for assessment of pesticide transport in fractured aquitards.

[5] The overall aim of the current study is to investigate water saturated vertical flow and solute transport through the 2 m thick nonoxidized zone of a fractured clayey till into a local aquifer (sandy layer) using a controlled field experiment at natural hydraulic gradient. Specific goals of the investigations are (1) to obtain field scale hydraulic conductivities and distribution of flow between the bulk till and fractures, (2) to evaluate the spatial distribution and influence of fracture aperture variability and root channels on total flow, (3) to obtain well defined field transport data of nonreactive tracer (bromide) and the pesticides mecoprop (MCP), metsulfuron and prochloraz, (4) to assess modeling strategies for application to solute transport in the combined till and local aquifer (sandy layer) of the experimental field. A specific goal for this modeling is to evaluate the suitability of using effective fracture spacings and mean apertures for conductive fractures derived from the field experiment in order to simulate concentrations observed in monitoring wells of the experimental field. It is an additional goal of the modeling to evaluate the suitability of using prior calibrated solute transport model input parameters obtained from the previous LUC experiments from the site by Jørgensen *et al.* [1998a].

2. Geological Setting and Field Site

[6] The site of the study is near the village of Havdrup, 40 km southwest of Copenhagen in Denmark, Figure 1. The site is located on a slightly undulating Weichselian till plain, which is approximately 5–16 m thick in the study area. The glacial sequence rests on sandy limestone and marl from the upper Paleocene, underlain by whitish Danian bryozoan limestone. The two types of limestone constitute the main aquifer of the region and are used for the water supply of Copenhagen.

[7] Wells and excavations at the site show a high degree of lateral uniformity in the glacial sequence. This is illustrated by a 0.3 to 1.0 m thick sandy layer, which occurs at 5 to 6 m depth over the entire site (approximately 10 hectares) and is also found in wells adjacent to the site (Figure 1). The upper 3 m of the till is seasonally water saturated, highly fractured, and oxidized to a yellowish brown color. Near-vertical fractures with prominent staining (primary fractures) are found throughout the oxidized zone and in some cases extend several meters down into the permanently water saturated gray nonoxidized clays. Most of the fractures in the upper few meters appear to be caused by desiccation during periods of low water table. Vertical repetition of desiccation fractures have also been observed deeper in the



- E3: Excavation for previous sampling of large undisturbed columns (Jørgensen *et al.* 1998a).
- E4: Experimental field of current investigation.
- Boring

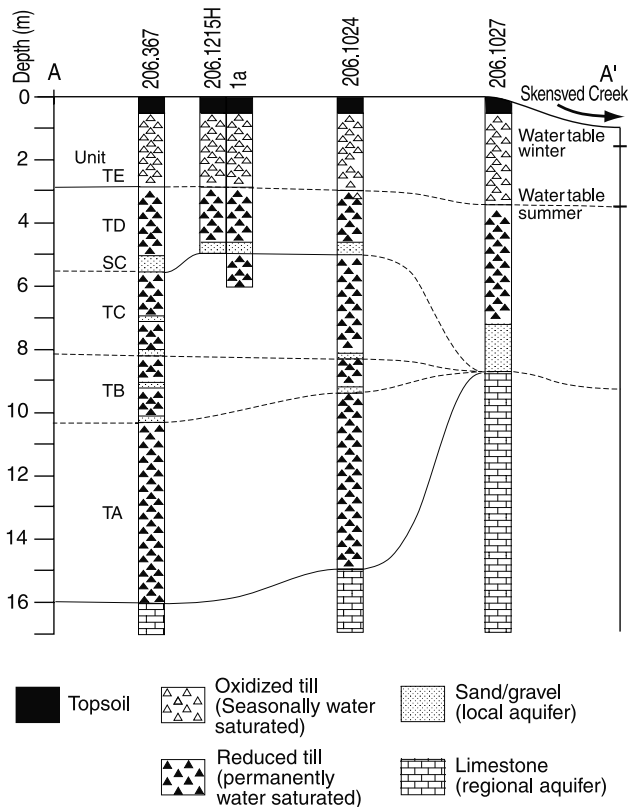


Figure 1. Location and geological profile of field site. See Jørgensen [1990] for explanation of stratigraphic units TA–TE.

till at the site, which suggest that fractures may not only have a general occurrence in the top of clay rich till deposits, but may also be present in older units deeper in glacial depositional sequences [Hoffmann, 2000]. The widely spaced near-vertical fractures in the nonoxidized zone appear to be due to glacial disturbance or tectonic movements in the bedrock. The latter origin is suggested by the similarity in strike

between the fractures of the till and an underlying regional fault separating the bryozoan and Paleocene limestone (Figure 1). Fracture apertures from the till have been determined using LUCs collected from the upper few meters of the till approximately 50 m north of the current experiment (Figure 1) [Jørgensen *et al.*, 1998a]. Flow and solute tracer transport experiments with these columns revealed collaborating mean hydraulic and transport fracture apertures decreasing from 84 μm at 1 m depth to 39–42 μm for fractures at 4 m depth. This distribution of apertures is in agreement with the earlier mentioned tendency for fracture apertures to decrease with depth as observed for other till sites in Denmark [Jørgensen *et al.*, 1998b].

3. Materials and Methods

3.1. Setup of Field Experiment

[8] In order to carry out flow and tracer transport experiments in the nonoxidized till, the oxidized zone was removed in a 17 m by 17 m wide and 3 m deep excavation (Figure 2). The base of the excavated field followed the transition between oxidized and nonoxidized till, which also represent the transition between seasonally and permanently water saturated till in the area (shown in Figure 1). Most of the experiments were conducted during summer and autumn where the natural water table in the till was at the same level as the bottom of the excavation. In the bottom of the excavation, two infiltration fields, each 4 by 10 meter long and 2 m deep were established. In vertical section each field included the till layer and a 0.3 m thick sandy layer (local aquifer) that formed a uniform natural base of the two fields. These fields are hereafter referred to as (field I) and (field II) and the sandy layer at the base of the fields is referred to as the base sand layer (BSL). In order to ensure that the fractures in the till were not obstructed by smearing from the excavation process, scrapers were used to remove the upper 3–5 cm of the fields, which were afterwards vacuumed for removal of loose particles. The two fields were separated by an impervious central wall oriented perpendicular to the primary fractures of the till. Parallel with this central wall two additional impervious walls were established along the outer edges of the total experimental field in order to hydraulically separate it from the surrounding till. The impervious walls were constructed by back filling the till material excavated from the wall trenches. Following the recommendations of Foged and Willie [1992] the back filled soil material was compacted with a hand worked soil compressor which ensured that the saturated hydraulic conductivity of the walls was less than 5×10^{-10} m/s. Dismantling the experimental fields after a dye tracer had been infiltrated at completion of all experiments indicated that there was no leakage along the walls. The bottom of each of the impervious walls was terminated at the upper boundary of the BSL. Water saturation and steady state lateral flow in the BSL during the experiment was controlled by constant injection of groundwater into a 10 m long injection tube, installed horizontally in the BSL underneath the central impervious wall. Hereby a groundwater divide was created in the BSL between the two infiltration fields. From this divide oppositely directed lateral groundwater flow would carry tracers (leached into the BSL from the above two till blocks) horizontally

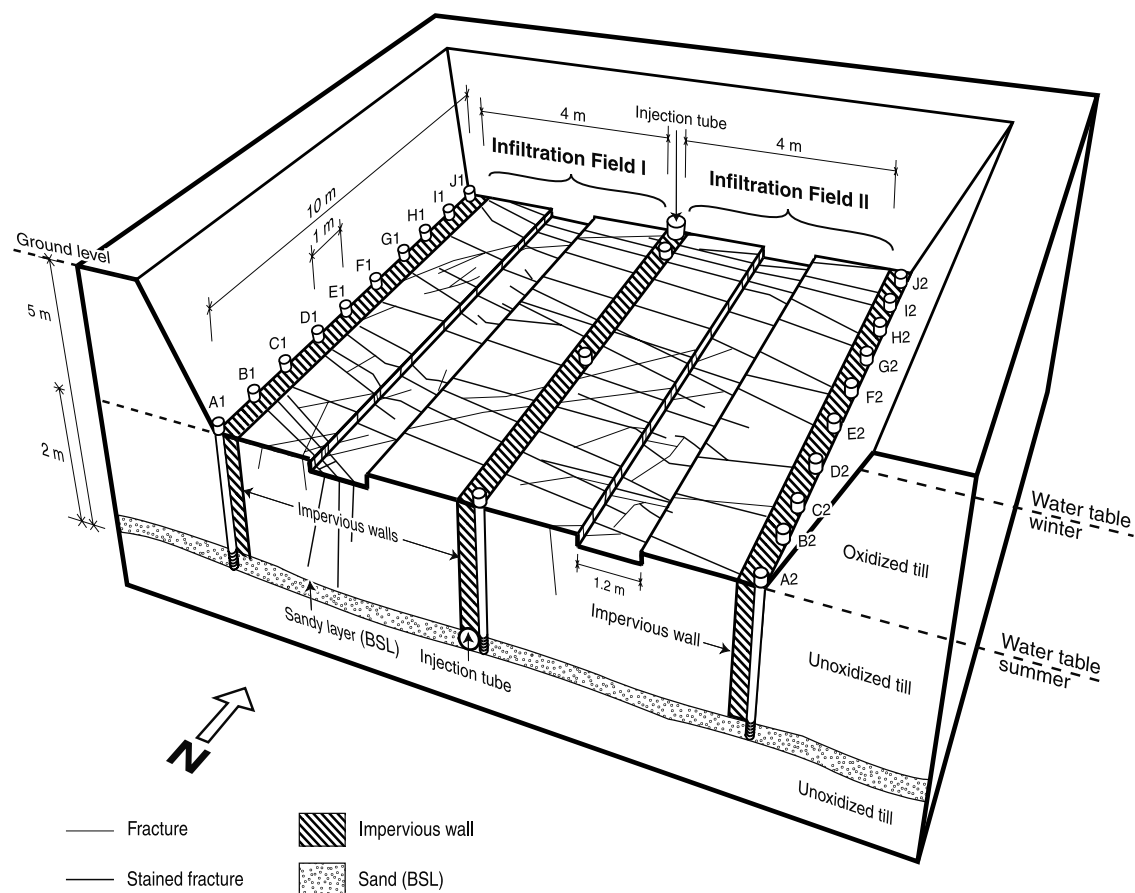


Figure 2. Field experimental setup showing infiltration fields I and II together with the observed distribution of secondary and primary (stained) fractures and location of monitoring wells installed in the sandy layer (BSL).

toward the monitoring wells, which had been installed as fences at the outside of each infiltration field (Figure 2). The groundwater injected in the BSL was extracted from the limestone aquifer at site. The monitoring wells in the BSL were screened over the total thickness of this layer and sealed with bentonite to the surface of the experimental field. Teflon lines was installed in each monitoring well and to reduce risk of tracer cross contamination between the wells an stationary sampling lines and lids which were fitting the sampling glass bottles directly was installed in each well. To avoid evaporation from the surface, the experimental fields were insulated and covered by impervious materials (Figure 3). During the experiments the influent solutions were keep near groundwater temperature (8° to 12°C) and the headspace above each of the two infiltration fields was air-conditioned. Further details of the experimental setup are shown in Figure 3.

[9] Fractures in the two experimental fields occurred with two predominant strikes (E-W and NNE-SSW), forming a primary and secondary set of mainly vertical fractures. These fractures are hereafter referred to as the primary and secondary fractures. The primary fractures were near perpendicular to the impervious walls and were easily distinguishable from the secondary fractures by a 4–20 cm wide reddish brown stained rim developed from natural chemical weathering of the gray nonoxidized clay along the fractures [Jørgensen and Fredericia, 1992]. Some

of the primary fractures connected to the under laying sand layer, while most of the secondary fractures terminated above the sand layer and had less horizontal length and less developed staining along the fracture rims. In addition to the near vertical fractures a few horizontal secondary fractures were observed in the upper 0.1–0.3 m of the nonoxidized till. The dip orientation, spacing and depth of the fractures observed in the experimental fields, are within the range of data for deep near vertical fractures observed from previous excavations at the site and other sites in Denmark [Jørgensen and Fredericia, 1992; Jørgensen *et al.*, 1998a, 1998b; Klint and Gravesen, 1999; McKay *et al.*, 1999]. The soil material in the two experimental fields appeared to be identical and furthermore resembled the large number of samples that have previously been collected from the same geological units at the study site and from elsewhere in Denmark. On this basis the experimental fields I and II are believed to be generally representative for fractures in the upper meters of nonoxidized clay-rich tills in the Danish area. In Table 1, the fracture spacing observed in each of the two infiltration fields are shown together with the saturated hydraulic conductivity of the unfractured soil material, total porosity, θ_m , and grain size distribution for the bulk till and BSL [European Testing Committee, 1995]. Further geological and mineralogical description of the soil material is given by Jørgensen and Fredericia [1992].

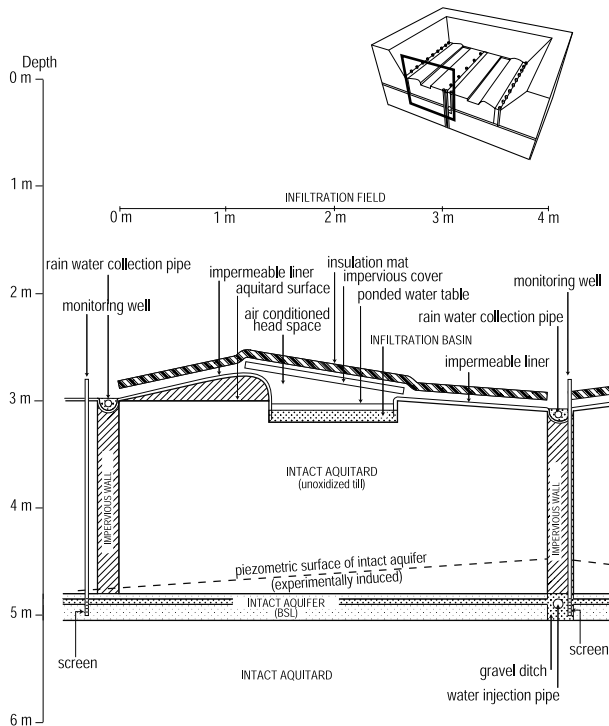


Figure 3. Vertical section through experimental field I showing experimental installations. Infiltration fields I and II are mirror images of one another.

3.2. Flow and Solute Transport Experiments

[10] Saturated vertical bulk hydraulic conductivity (K) for each of the two infiltration fields was determined from the falling head of the water table in each of the infiltration fields (Figure 3). Evaporation was measured from evaporation pans installed in the basins. No evaporation, however, was observed from the pans, which show that the setup was tight. The vertical hydraulic gradient (0.8), which was used in all experiments, represent the lower value of natural gradients (0.8–2.3) previously observed from the same depth in the till by Jørgensen and Spliid [1998]. The gradient in the experiments was controlled by the elevation difference between a constant head in the infiltration fields and the constant head in the BSL generated by the steady state groundwater

injection (40 L/h) in this layer (Figure 3). All flow measurements and solute transport experiments were carried out after water saturation and steady state flow through the total experimental field had been established. Solute transport experiments were carried out using pulses of bromide (conservative tracer) applied to field I and II, and a mixed pesticide pulse applied to field I before the bromide experiments. The length of the bromide and pesticide pulse and afterward “flushing” together with applied concentrations of compounds are shown in Table 2. The applied pesticides were mecoprop (MCP) and metsulfuron-methyl (mobile contaminants), and prochloraz (immobile contaminant). Equilibrium partition coefficients (K_d -values) for MCP, metsulfuron-methyl and prochloraz in soils and sub-soils are 0.07–0.86 [Helweg and Fomsgaard, 1996], <0.01–0.19 [Lindhard et al., 2000] and 500–800 [Tomlin, 1994], respectively. After completion of the tracer experiments the dye tracers “Na-Flourescien” and “brilliant blue” was applied to field I and II in order to trace the flow paths. In field I, tracing of the complete length of flow paths from the infiltration basin to the monitoring wells was achieved by application of the dye tracers until clear breakthrough was observed in the monitoring wells (Figure 2). In field II, the flow injection into the sandy layer (Figure 2) was stopped before the dye was applied in order to enable study of the arrival of tracer at the interface between the lower boundary of the till and the top of the BSL. The Na-Flourescien behaves similar to a conservative tracer in the till, while the Brilliant Blue is slightly retarded [Helstrup, 1998]. Concentrations of compounds in the individual experiments are shown in Table 2.

3.3. Mapping of Fractures and Flow Paths

[11] After completion of all experiments the two infiltration fields were dissected into a horizontal section 0.15 m below the surface of the infiltration fields, and two vertical sections along the infiltration fields. A 3D coordinate grid was established for the total plot, and along the dissected sections the fracture strike, dip, width and depth was measured, together with the distribution of the dye tracer.

3.4. Large Undisturbed Column Experiments

[12] Three undisturbed columns (LUC) each 0.5 m in diameter by 0.5 m high were collected from the experimental field I. The columns were collected from 4.5–5 m depth representing the lower 0.5 m of the till in the experimental

Table 1. Physical and Textural Properties of Till Material and Sand Layer (BSL)

Infiltration Field	Depth, m	Clay/Silt/Sand/Gravel, ^a %	Total Porosity, ^a %	Hydraulic Conductivity of Unfractured Soil Sediment, m/s	Fracture Spacing 2B, ^b m			
					Primary Fractures		Secondary Fractures	
					3 m Depth	5 m Depth	3 m Depth	5 m Depth
I	3–5	21–28/33–41/25–42/4–6	0.25–0.27	3.2×10^{-10} ^a	0.52	1.25	0.77	5.41
II	3–5	—	0.25–0.27	—	0.61	1.1	2.5	5.0
Base sand layer (BSL) fields I and II	5–5.5	15–40 [clay and silt]/65–70/20–7 ^c	—	2.8×10^{-4} ^d	—	—	—	—

^a Determined from core samples collected from infiltration field I.

^b Calculated from $2B = (\text{measured fracture width on horizontal surface/area of horizontal surface})^{-1}$. Fracture width is defined as the measured fracture length on horizontal surfaces.

^c Mean values of upward coarsening unit.

^d Representing the upper 5 cm gravel of the upward coarsening unit [Hoffmann, 2000].

Table 2. Solute Tracers Applied in Field Experiment

Infiltration Field	Infiltrated Volumes, L		Infiltrated Concentrations, mg/L					
	Bromide(pulse) [flushing]	Pesticide(pulse) [flushing]	Bromide	MCPP	Metsulfuron	Prochloraz	Brilliant Blue	Na-Flourescien
I	(400)[1000]	(400)[1800]	4000	12.00	0.16	4.50	10000	3300
II	(400)[325]	—	3870–1850	—	—	—	10000	3300

field (Figure 1). Each column contained one pronounced vertical primary fracture, which clearly connected through the infiltration field to the BSL below the field. One of the columns (III) was collected along a clearly dyed fracture flow path, while the fracture of the other two columns (I and II) did not contain dye in visible concentrations. The columns were excavated with hand tools to prevent disturbance. After excavation, but before being disconnected from the geolo-gical formation at their base, the columns were covered with a flexible rubber membrane and encapsulated in a stabilizing transport metal casing. In the laboratory the columns were installed in a flexible wall triaxial setup with a confining pressure of 80 KPa and temperature of 10 C°. Bulk hydraulic conductivity values for the saturated columns were measured by recording effluent yield from the columns at constant hydraulic gradient (0.8–1). The hydraulic gradient was controlled by the elevation difference between a constant head influent reservoir and the end of the effluent tubing. The laboratory experimental setup is shown in Figure 4. Further description of the LUC procedure is given by Jørgensen *et al.* [1998a].

3.5. Chemical Analysis

[13] Water samples collected from the pesticide experiment was extracted for metsulfuron-methyl, MCPP and prochloraz. The samples were extracted immediately after sampling or kept at 4°C until analysis, which was not later than 24 hours after sampling. Samples (1 L) were extracted by solid phase extraction on Porapak RDX cartridges (Waters). Metsulfuron-methyl and MCPP were analyzed by Liquid Chromatography (LC) with ultraviolet detection (UV) at a wavelength of 229 nm. LC with mass spectrometric (MS) detection was employed for verification of positive samples containing metsulfuron-methyl and MCPP and for the analysis of prochloraz. Metsulfuron-methyl and MCPP were analyzed by electro spray ionization (ESI) in negative ionization mode, while prochloraz was analyzed by atmospheric pressure chemical ionization (APCI) in positive ionization mode. For further details of extraction method and LC/MS analysis see *Spliid and Køppen* [1996] and *Køppen and Spliid* [1998]. Analysis for bromide in water samples collected from the monitoring wells was carried out using a selectrode (Radiometer A/S). The detection limit for the selectrode was 1 mg/L.

3.6. Transport Model Code (FRAC3Dvs)

[14] A Discrete Fracture Matrix Diffusion (DFMD) numerical 3D-model was used for the flow and pesticide transport simulations. FRAC3Dvs, version 4.0 is developed for multiphase steady state groundwater flow and contaminant transport in porous or discretely fractured formations [Therrien and Sudicky, 1996]. The model determines flow rates in the matrix using Darcy's Law, based on the

hydraulic conductivity (K), the hydraulic gradient (i) and the matrix porosity (θ_m). Fractures are represented as uniform openings between parallel plates and flow velocities in the fractures are determined with the cubic law (see Table 5). Fracture spacing ($2B$) and aperture ($2b$) values are specified as part of the nodal input and may be varied in orthogonal network patterns in the 3D framework. FRAC3Dvs assigns advective-dispersive equations to both the fractures and the matrix and allows transfer of solutes between fractures and matrix. The FRAC3Dvs model can also simulate single porosity (SP) continuum medium flow and solute transport by not defining fractures. This setup of the model was used for simulation of flow and transport in the BSL layer in the experiment. For chemically reactive contaminants, the model assumes that sorption is a linear equilibrium process independent of sorption capacity of the medium, flow velocity or residence time. Sorption is described by use of the retardation factor for the fractures (R_f) and for the matrix (R). These are required input parameters together with values for effective diffusion coefficients (D_e) for solutes in the matrix and free water diffusion coefficient (D_w). Loss of degradable solutes is described by a first-order decay constant (λ). Hydrodynamic dispersion in the fractures is simulated using the free water diffusion coefficient and an input value of longitudinal dispersivity for the fractures (α_{l*}). Input values of longi-

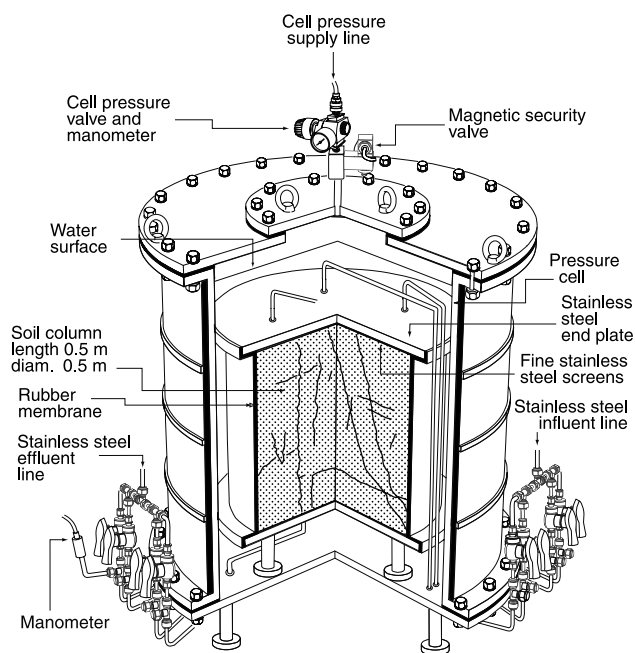


Figure 4. Flexible wall permeameter for large undisturbed column (LUC) laboratory experiments.

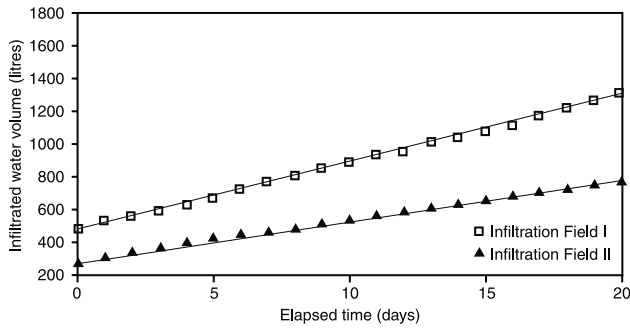


Figure 5. Cumulated flow observed for experimental fields I and II.

tudinal and transverse dispersivity (α_l and α_t) are also needed to determine hydrodynamic dispersion in the matrix.

4. Results

4.1. Flow and Transport Investigations

4.1.1. Bulk Flow Experiments

[15] The observed flow for each of the two infiltration fields is shown in Figure 5, and the corresponding bulk hydraulic conductivities are given in Table 3. The straight-line relationship observed between flow (accumulated infiltration) and time for the two infiltration fields show that steady state flow conditions were obtained for both fields. This shows that the flow system was physically stable. Calculated from the Darcy equation, the saturated hydraulic conductivity for the two experimental fields was 8.5×10^{-9} and 15×10^{-9} m/s, respectively. These hydraulic conductivities are approximately two orders of magnitude higher than the hydraulic conductivity for unfractured clay samples (Table 1). Estimated from these hydraulic conductivities, 98% and 96% of total flow in field I and field II, respectively, was conducted along preferential flow paths in the till. The small difference in bulk hydraulic conductivity observed between the two experimental fields show that the influence of preferential flow was similar in the two fields and suggest that the fields were representative experimental volumes (REV) for the till at this depth. The hydraulic conductivity corroborate furthermore with K-values reported from other nonoxidized clayey till aquitards ($10 \times 10^{-6} - 10 \times 10^{-8}$ m/s) in Denmark [Fredericia, 1990].

4.1.2. Bromide and Pesticide Breakthrough

[16] Normalized breakthrough curves obtained for bromide from each monitoring well in the two experimental fields are shown in Figure 6. In field I, breakthrough of bromide occurred in 5 out of 10 wells. High-normalized concentrations were observed in the wells A1, D1 and E1. The first arrival occurred 3 to 5 days after tracer application,

and maximum normalized concentrations of 0.025 to 0.3 occurred after 12 to 15 days. Distinctively later arriving bromide was observed from monitoring wells B1 and C1 with breakthrough concentrations of only 0.8% to 5% relative to the maximum peak concentrations. In field II bromide was only detected in only one well (C2) out of 10 wells. The migration rate observed in C2 was similar to those observed in field I (Figure 6a). However, the maximum breakthrough concentration was similar to the low concentrations recorded in field I, and hence, the very high relative breakthrough concentrations observed in field I was apparently absent from field II.

[17] Figure 7 shows the normalized breakthrough of a pesticide pulse (mecoprop (MCP), metsulfuron, and prochloraz) applied to field I. Distribution of high and low breakthrough concentrations for MCP and metsulfuron in the wells was the same as observed for bromide. In the wells with high breakthrough concentrations the migration velocities and normalized concentrations for the pesticides and bromide were approximately the same, which indicate that retardation because of degradation of the pesticides was small within the short timescale of the experiments. In the wells with low breakthrough, however, the breakthrough concentrations were approximately 1 order of magnitude lower for the pesticide than for the bromide. The front of

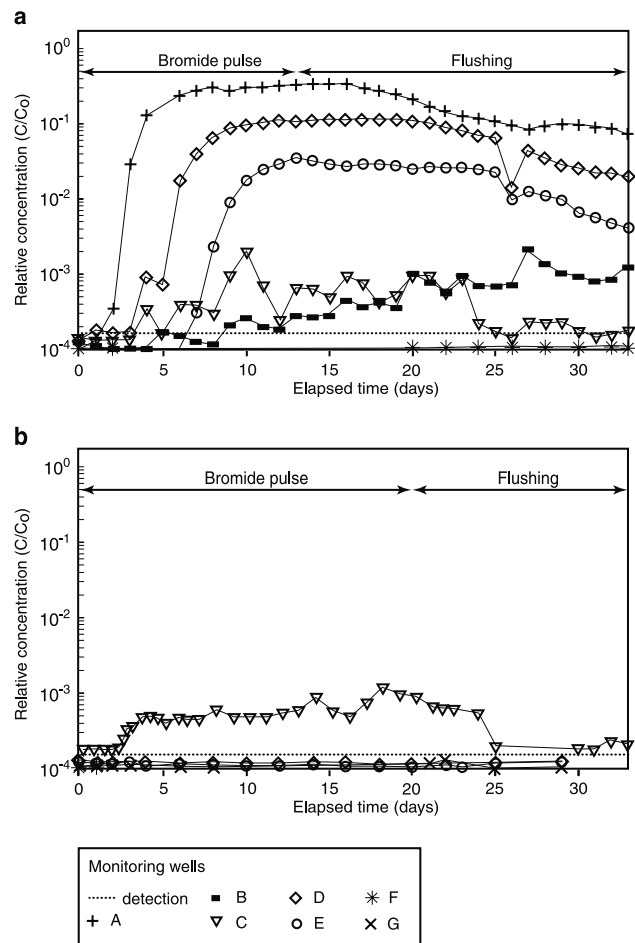


Table 3. Bulk Hydraulic Properties Determined for the Infiltration Fields (Figure 2)

Infiltration Field	Applied Hydraulic Gradient i	Flow Rates, QL/d	Hydraulic Conductivity K, ^a m/s
I	0.8	41.5 ^a	15×10^{-9}
II	0.8	24 ^a	8.5×10^{-9}

^aQ = KA*i*, where A = 40 m² (area of infiltration field).

Figure 6. Breakthrough of bromide in the monitoring wells of experimental fields (a) I and (b) II. Location of monitoring wells is shown in Figure 8.

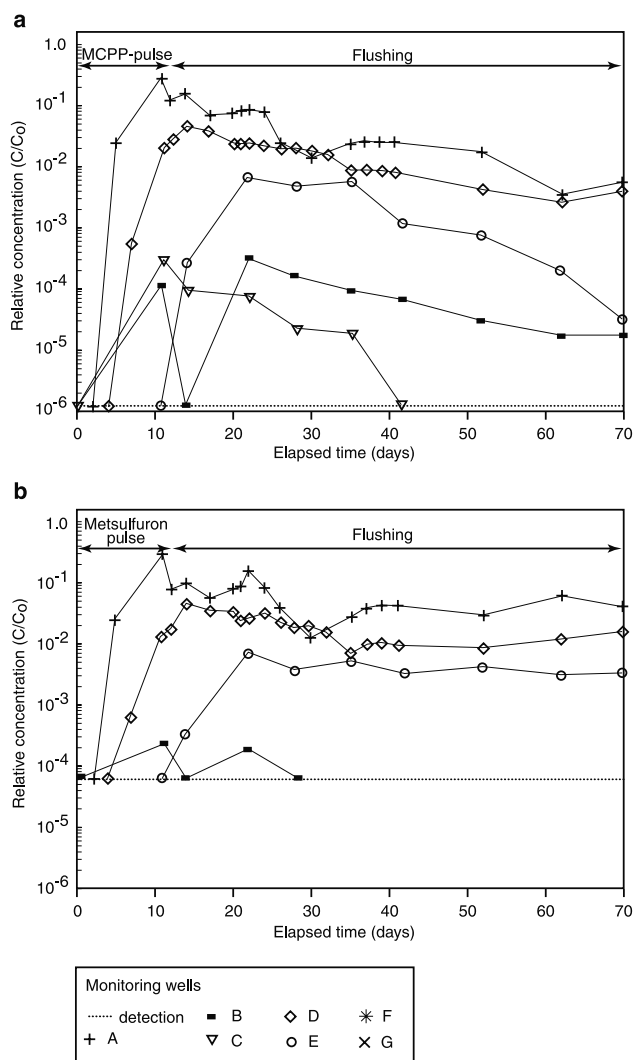


Figure 7. Breakthrough of the herbicides mecoprop (MCPP) and metsulfuron in the monitoring wells of experimental field I. Location of the monitoring wells are shown in Figure 8.

both the bromide and pesticides migrated at several m/d and arrived at pore volumes (PV) much less than 1 PV, which is indicating a very high degree of preferential flow. Prochloraz was identified only in single samples in concentrations near the detection limit. This is likely due to the strong sorption of this pesticide in the till and BSL. The overall control of preferential flow processes on flow and solute transport in the field experiment is in agreement with studies using large undisturbed columns from previous column studies from the current site and other clayey till sites in Denmark [Jørgensen *et al.*, 1998a, 1998b, 1998c].

4.1.3. Dye Tracer Experiment and Fracture Aperture Variability

[18] After completion of all experiments, the flow paths in the experimental fields were investigated by infiltrating dye tracers in both experimental fields, followed by excavation of the fields (Figure 8). Excavation of horizontal sections in the two experimental fields revealed that the dye occurred along distinct segments of the primary fractures in the till, while it was absent from the secondary and few horizontal

fractures and the unfractured till matrix (Figure 8a). Measured on horizontal sections at 3 and 5 m depth in the experimental fields, the dyed fracture sections constituted only 9.8 to 26.1%, respectively, of the total observed horizontal length (fracture width in Table 4). In consequence the effective fracture spacing (Table 4) at the base of the till in the experimental fields was 4 to 5 m, which is approximately 4 to 5 times greater than the observed spacing between stained fractures (Table 1).

[19] Most of the dyed fracture sections could furthermore be subdivided into 0.02–0.1 m wide channels, which formed near vertical flow paths, some of which connected to the underlying sandy layer at 5 m depth (Figure 8b). The flow paths in the till were inspected both from exposed fracture surfaces, and on the upper boundary of the sandy layer after removal of the overlying till. On the fracture surfaces the dye formed a distinct fingering pattern resembling root channels. The root channels were found to conduct the dye tracer along all the inspected fractures, which suggests that fracture aperture enhancements due to tectonics or desiccation were less important. The channels were occasionally identified as physical channels (root casts), while actual root remains in the channels were not found. This suggests that the channels may be very old, potentially relating back to periods where Denmark to a high extent was covered by forest (more than thousand years). The observed root channels with active flow resemble the channel features described previously from the same site by Jørgensen [1990]. On the upper surface of the BSL the dye was distributed as isolated spots along the lines where the fractures had been intersecting with the BSL (Figure 9). In agreement with the observed root pattern along the exposed fracture surfaces these spots resembled outlets from root channels penetrating into the BSL. The finding of deep hydraulically conductive root channels is in agreement with observations from other Danish tills and suggests that channels from tree roots may play an important role for water flow to at least 5–6 m depth in the clay-rich tills.

[20] The distribution of flow along dyed and nondyed fracture sections in the till was further evaluated from laboratory flow experiments using large undisturbed columns (LUC) collected from field I (Figure 10). Two columns were collected along prominent, single, near-vertical, nondyed fractures connecting to the sand layer at $x = 1.5$ m and 7.8 m, respectively (section B-B' in Figure 8b). A third column (III) was collected along a dyed fracture section and contained a 0.05 m wide dyed fracture flow channel that was connecting to the sand layer at $x = 4$ m in section B-B' (Figure 8b). The flow experiments revealed that the columns (I and II) collected from the nondyed fracture sections had 2 orders of magnitude lower hydraulic conductivity than the column (III) taken from the dyed fracture section (Table 5). Based on the hydraulic conductivity and using the "cubic law", the calculated hydraulic aperture of the fracture in columns I and II was 12 μm and 14 μm , respectively, while the aperture for the dyed fracture flow channel of column III was 124 μm (Table 5). Flow velocity in the channel was 830 m/d to 2.3 km/d calculated for the range of natural hydraulic gradients observed in the till, while flow rates in the nondyed fracture section was 9–22 m/d calculated for the same hydraulic gradients (1 to 2.3 [Jørgensen and Spliid, 1998]). The absence of dye tracer in the fracture regions represented

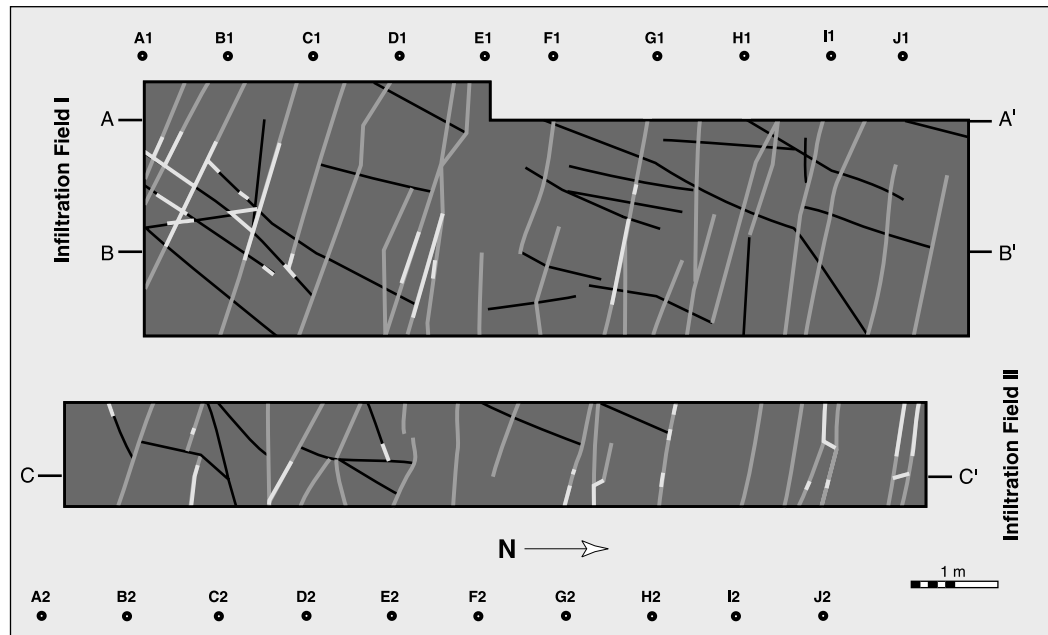
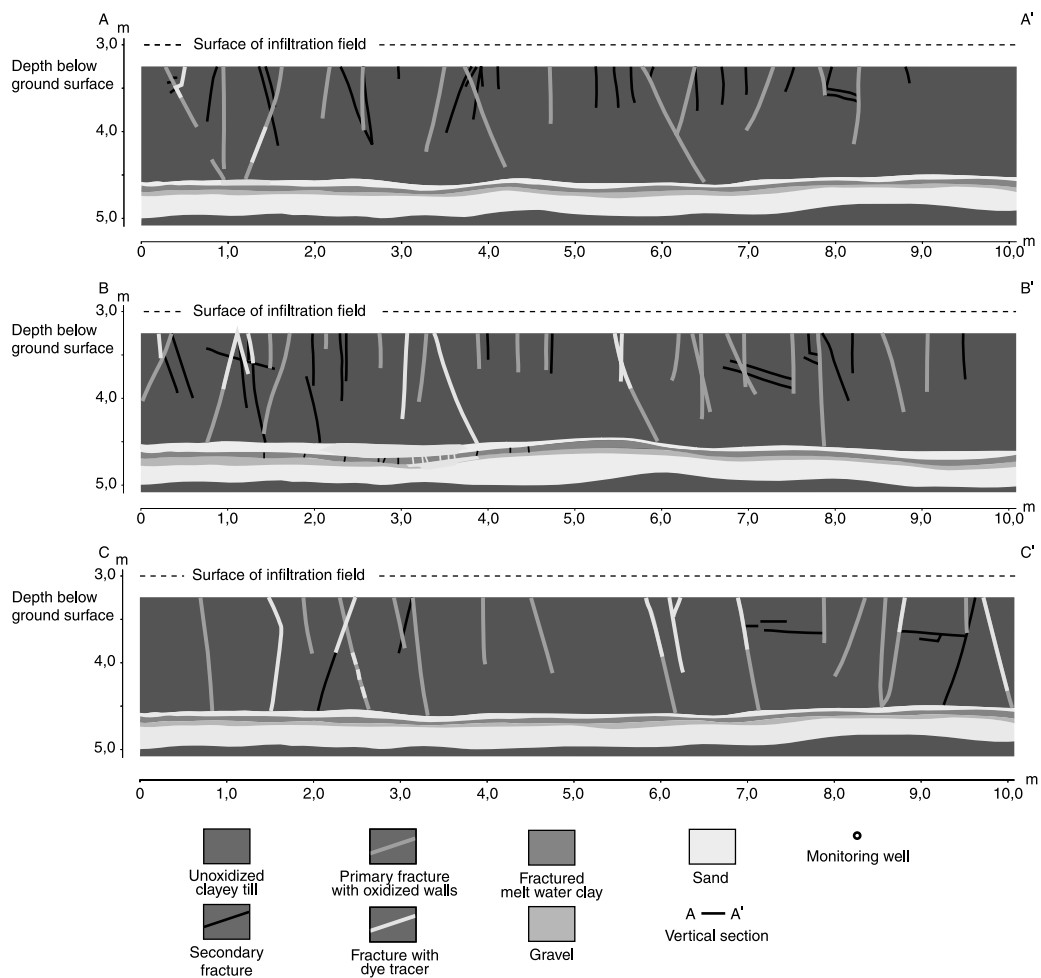
a: Horizontal section**b: Vertical sections**

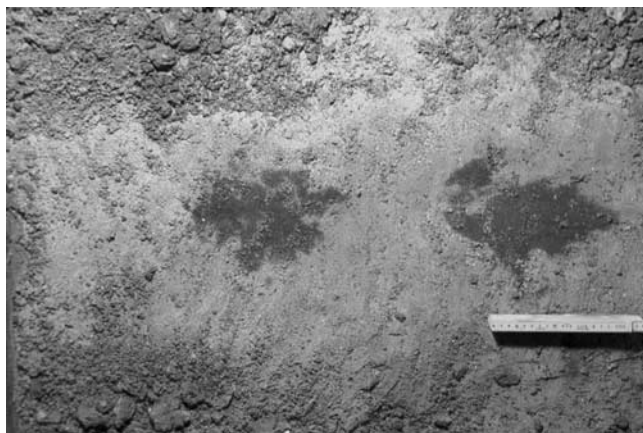
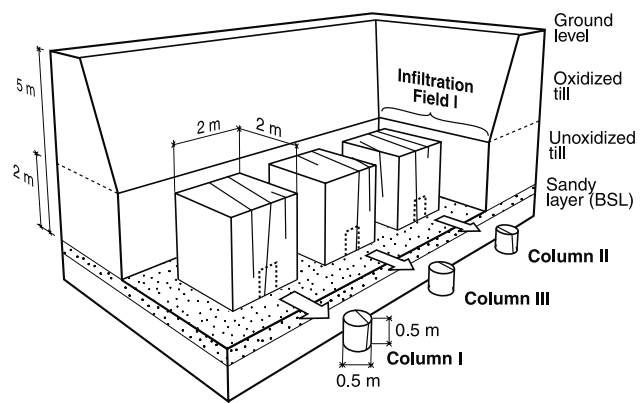
Figure 8. Distribution of fractures and dyed flow paths in (a) horizontal and (b) vertical sections of experimental fields I and II. See color version of this figure at back of this issue.

Table 4. Frequency and Spacing of Dyed Fractures at 3 m and 5 m Depth in the Two Infiltration Fields (Figure 2)

Infiltration Field	Total Width of Secondary plus Primary Fractures, m/m ²		Width of Dyed Fractures, % of Total Width		Spacing of Dyed Fractures, 2B Effective, ^a m	
	3 m Depth	5 m Depth	3 m Depth	5 m Depth	3 m Depth	5 m Depth
I	3.22	0.98	9.8	19.7	3.2	5.17
II	2.59	0.95	15.7	26.1	2.47	4.0

^a 2B = (width of dyed fractures/area of measurement)⁻¹. Fracture width is defined as the measured fracture length on horizontal surfaces.

by columns I and II is consistent with the small fracture apertures of these columns and the relative short period of the dye infiltration in the field experiment (9 days). Using the mean aperture value 13 μm (columns I and II) for the nondyed fractures, a mean aperture of 49 μm was calculated for the dyed fractures in experimental field I. The value was calculated on the basis of the cubic law using the bulk and matrix hydraulic conductivities given in Table 1, and the measured lengths of dyed and nondyed fracture sections at 5 m depth of field I (Table 4). On the basis of the mean aperture values 13 μm and 49 μm , respectively, for the dyed and nondyed fractures, and their corresponding widths, it was estimated that 92% of total bulk flow occurred along the dyed fracture root channels, while only 6% of bulk flow occurred along the nondyed fractures in experimental field I. The great range of fracture apertures (12–124 μm) obtained from the LUCs show that individual LUCs should not be expected to generally represent mean field aperture values. Hydraulic parameters and distribution of flow for fractures and till matrix obtained from field I is summarized in Table 6.

**Figure 9.** Distribution of dye tracer on exposed horizontal surface representing the boundary between the till and the sandy layer (BSL) in experimental field II. The spot-wise appearance of the dye correlates with fracture root channels observed in the overriding till. Folded yardstick (right in the picture) is 0.1 m long. See color version of this figure at back of this issue.**Figure 10.** Sampling of large undisturbed columns (LUC) from field I after flow and tracer experiments.

4.1.4. Correlation Between Solute Breakthrough and Dyed Flow Paths

[21] In Figure 11 the dyed flow paths in the till and the BSL are correlated with the solute maximum breakthrough concentrations observed in the monitoring wells. This reveals that the breakthrough of the pesticides and bromide in monitoring screens A1 to E1 correlates with the dyed flow paths in both the till and the BSL. From the distribution of dye in the BSL it is suggested that the low bromide and pesticide concentrations observed in wells B and C represents the dispersed flanks of the pesticide “plumes” observed in well A and D rather than transport from fractures with lesser aperture. Transport of bromide, pesticide or dye was not observed down gradient from the fractures with LUC hydraulic apertures of 12–14 μm (Table 5).

[22] As mentioned earlier the application of the dye tracer into field II, occurred after the injected water flow in the BSL was stopped, and hence, no dyed traces of lateral transport in the sandy layer toward the monitoring wells was available from this field. However, there is correlation between the single finding of the bromide (well C2) in field II, and the up gradient position of the only fracture, that was dyed into the sandy layer (see Figures 8a and 8b). As mentioned before the breakthrough concentration observed from this well is significantly lower than the maximum concentrations observed from the dyed flow

Table 5. Hydraulic Properties and Calculated Hydraulic Apertures of Large Undisturbed Columns (LUC) Collected From Infiltration Field I

Large Undisturbed Column (LUC)	Bulk Hydraulic Conductivity, (m/s)	Bulk Flow Rates Q , ^a (m/d)	Total Width of Fracture [Width of Dyed Section], m	Hydraulic Fracture Aperture, ^b μm
I [nondyed]	4.4×10^{-9}	3.8×10^{-4}	0.5 [0.0]	14
II [nondyed]	3.0×10^{-9}	2.6×10^{-4}	0.5 [0.0]	12
III [dyed fractures]	1.2×10^{-7}	1.0×10^{-2}	0.5 [0.05]	124

^a Gradient in LUC, $i = 1.0$.

^b Parallel fracture system: $2b = [(Q/12\mu)/(WNipg)]^{1/3}$ [Snow, 1968], where $2b$ is the fracture aperture (μm), μ is the viscosity of water (kg/m s), ρ is the density of water (kg/m^3), g is the gravitational constant (m/s^2), W is the fracture width (trace length) on top of column (m), N is the number of fractures, i is the gradient across the column, and Q is the volumetric flow rate (m^3/s).

Table 6. Summary of Discharge and Flow Paths Determined for Infiltration Field I^a

Flow Domain	Percentage of Total Volume, %	Percentage of Total Flow, %	Mean Hydraulic Aperture, μm	Hydraulic Conductivity, m/s	Mean Flow Rate at Natural Hydraulic Gradients, ^b m/d
Bulk till	100	100	—	1.5×10^{-8} ^c	$1-3 \times 10^{-3}$
Nonfractured till	>99.9	2	—	3.2×10^{-10} ^c	$3-7 \times 10^{-5}$
Nondyed fractures	7.9×10^{-4}	6	13 ^d	1.1×10^{-4} ^d	9–22
Dyed fractures ^e	9.4×10^{-4}	92	49	1.4×10^{-3}	128–307

^a See text for explanation of calculations.^b See footnote a; $i = 0.8-2.3$ [Jørgensen and Spliid, 1998].^c Measured value.^d Calculated mean value from the large undisturbed columns I and II (Table 5).^e Mean values calculated from the “cubic law” using fixed values for the hydraulic conductivities for the bulk till, (see footnote a: 1.5×10^{-8} m/s) and the nonfractured till matrix, ($^{(1)}3.2 \times 10^{-10}$ m/s), and the mean LUC aperture ($^{(1)}13\mu\text{m}$) and width ($^{(1)}15.6$ m) for the nondyed primary fractures at 5 m depth.

paths in field I. In both experimental fields channeling in the fractures were clearly present as indicated by the dye tracer experiment, however, the difference in bromide transport between the two fields, may suggest that flow was more influenced by large aperture fracture channeling in field I than in field II. However, the observed low concentrations may equally well be reflecting the position of the well in the flank of a bromide plume that has bypassed the well.

4.2. Numerical Modeling

[23] The bromide and pesticide concentrations in the monitoring wells were simulated using FRAC3Dvs in the DFMD mode and SP mode for modeling of the transport in the till layer and the BSL, respectively. The simulations were carried out for the two monitoring wells A1 and D1 in field 1, because these wells appeared to represent well-

defined observation points in the core of the two separate plumes emanating from the conductive fracture sections in the till (see Figure 11).

[24] In order to assess modeling strategies for application to the observed solute transport, the simulations evaluates the suitability of using the simple mean fracture spacings (2B effective, Table 4) and aperture (dyed fractures, Table 6) values derived for the till as model input parameters, as oppose to data for the large number of individual fracture channels, which would be very difficult to obtain. In accordance, it was assumed that the fractures in the till had uniform apertures throughout the experimental field (Figure 12) and that that the spacing between these fractures was the same as the spacing between the fracture channel sections connecting with the aquifer at 5 m depth, (2B effective Table 4). The simulations were fit to the solute breakthrough

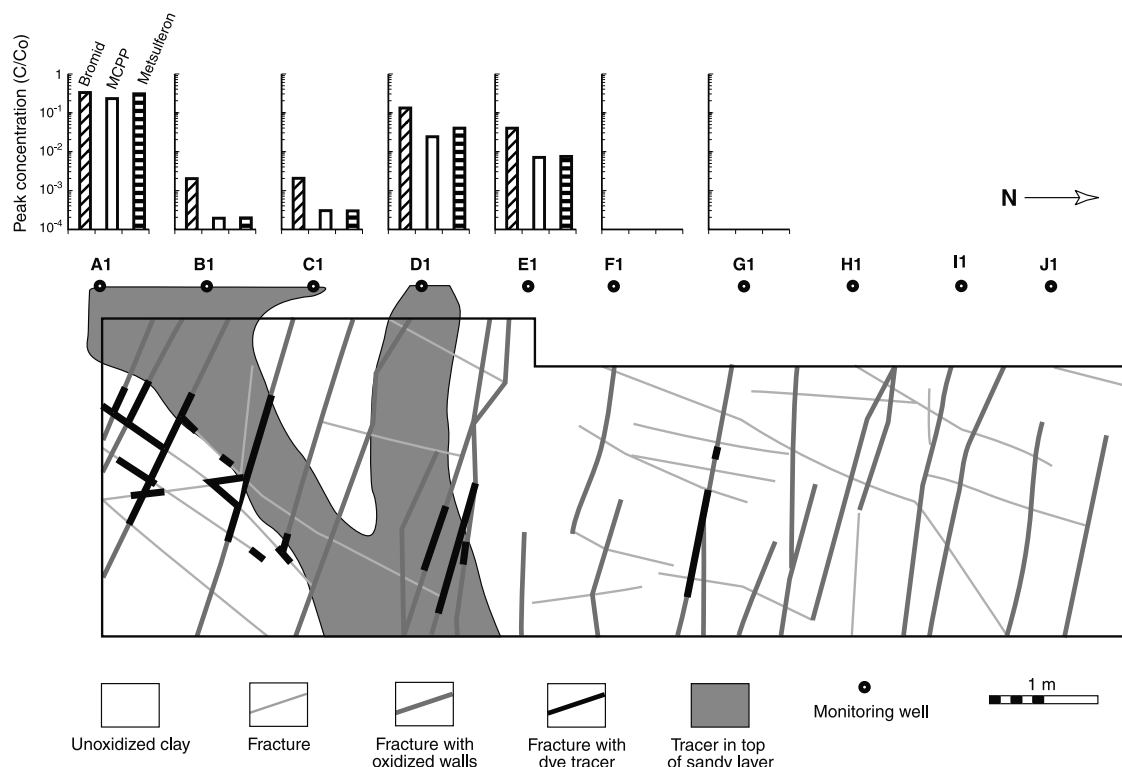


Figure 11. Correlation between dyed flow paths and maximum breakthrough concentration of bromide, MCP, and metsulfuron in monitoring wells in infiltration field I. See color version of this figure at back of this issue.

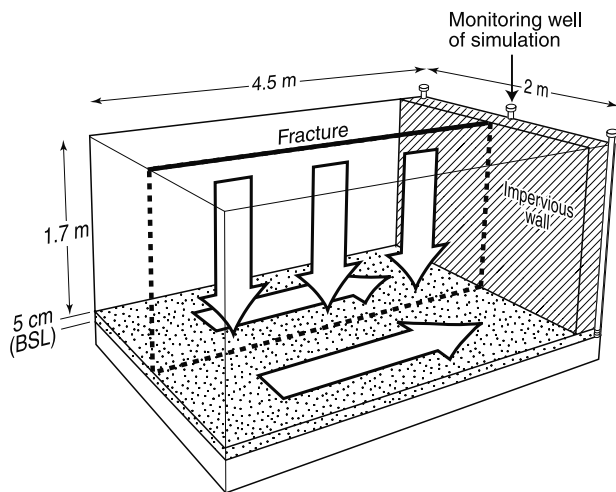


Figure 12. Conceptual fracture and flow model used for FRAC3Dvs simulations of bromide and pesticide breakthrough in monitoring well A1 and D1 (arrows denotes direction of flow).

observed from the wells by varying the fracture aperture value as the only fitting parameter. The evaluation of suitability of the approach was based on comparison between the fitted transport aperture value and the measured mean hydraulic aperture value for the conductive fracture

sections derived from the experiment (49 μm , Table 6). Porous medium and solute model input data used in the simulations are shown in Table 7.

[25] The simulated breakthrough of bromide, MCPP and metsulfuron is shown in Figure 13 and reveals that the simulations can be reasonably fitted to the observed data, when using fracture transport apertures of 60 μm and 40 μm for the fractures located up-gradient to wells A1 and D1, respectively. The result suggest that using the hydraulic fracture aperture mean value of 49 μm coupled with the calculated (effective) spacing for the dyed fracture flow paths (Table 4) the modeling would provide a reasonable approximation to both the observed flow and the concentrations of bromide and pesticides in the wells. The simulation (not shown) taking into account the observed 5 cm wide fracture channel (plume connecting with well D) slightly improved the fit to the breakthrough data for the well. The apparent similarity between the channel simulations and the uniform aperture fracture simulation is due to the location of the fracture channel in the middle of the experimental field. This involves a longer simulated transport path along the BSL to the monitoring well for the channel simulation, than for the case with uniform aperture fracture (terminating very near to the monitoring well, Figure 12). Hence, the quicker and stronger breakthrough of the solutes from the fracture channel into the BSL is apparently counteracted by more dispersion and retardation along the longer transport path in the BSL to the monitoring well. The small difference in the

Table 7. Soil and Solute Model Input Parameters Used in FRAC3Dvs Simulations

Parameter	Value	Source
Hydraulic conductivity K, m/s		
Till matrix	3.2×10^{-10}	(Table 1)
Sand layer (BSL)	2.8×10^{-4}	(Table 1)
Gradient, i		
Till	0.8	measured
BSL	0.035	measured
Dispersivity, $\alpha_L/\alpha_{TH}/\alpha_{TV}$, m		
Till matrix	0.1 / 0.01 / 0.01	Jørgensen et al. [1998a]
BSL	A1: 0.5 / 0.0167 / 0.00167	Hoffmann [2000]
BSL	D1: 0.08 / 0.0267 / 0.00267	Hoffmann [2000]
Fracture dispersivity α_L , m	0.1	Jørgensen et al. [1998a]
Porosity Θ_m		
Till matrix	0.25	measured
BSL	0.35	assumed
Fluid Viscosity μ , kg/m s		
8°C	1.386×10^{-3}	Lide [1994]
10°C	1.307×10^{-3}	Lide [1994]
Fluid density ρ , kg/m ³		
8°C	999.9	Lide [1994]
10°C	999.7	Lide [1994]
Free water diffusion coefficient D_w , ^a m ² /s		
Bromide (8°C)	1.96×10^{-9}	Lide [1994]
MCPP (10°C)	3.9×10^{-10}	Jørgensen et al. [1998a]
Metsulfuron (10°C)	3.34×10^{-10}	Worch [1993]
Retardation factor R^b		
Till		
Bromide	1.0	assumed
MCPP	6.8	Jørgensen et al. [1998a]
Metsulfuron	6.8	assumed
BSL		
Bromide	1.0	assumed
MCPP	1.0	Lindhard et al. [2000]
Metsulfuron	1.0	Lindhard et al. [2000]

^a The effective diffusion coefficient, $D^* = D_w \cdot \tau$, where τ is assumed equal to the matrix porosity $[\Theta_m]$ (Table 1).

^b R_{matrix} and $R_{\text{fracture wall}}$ are assumed equal values.

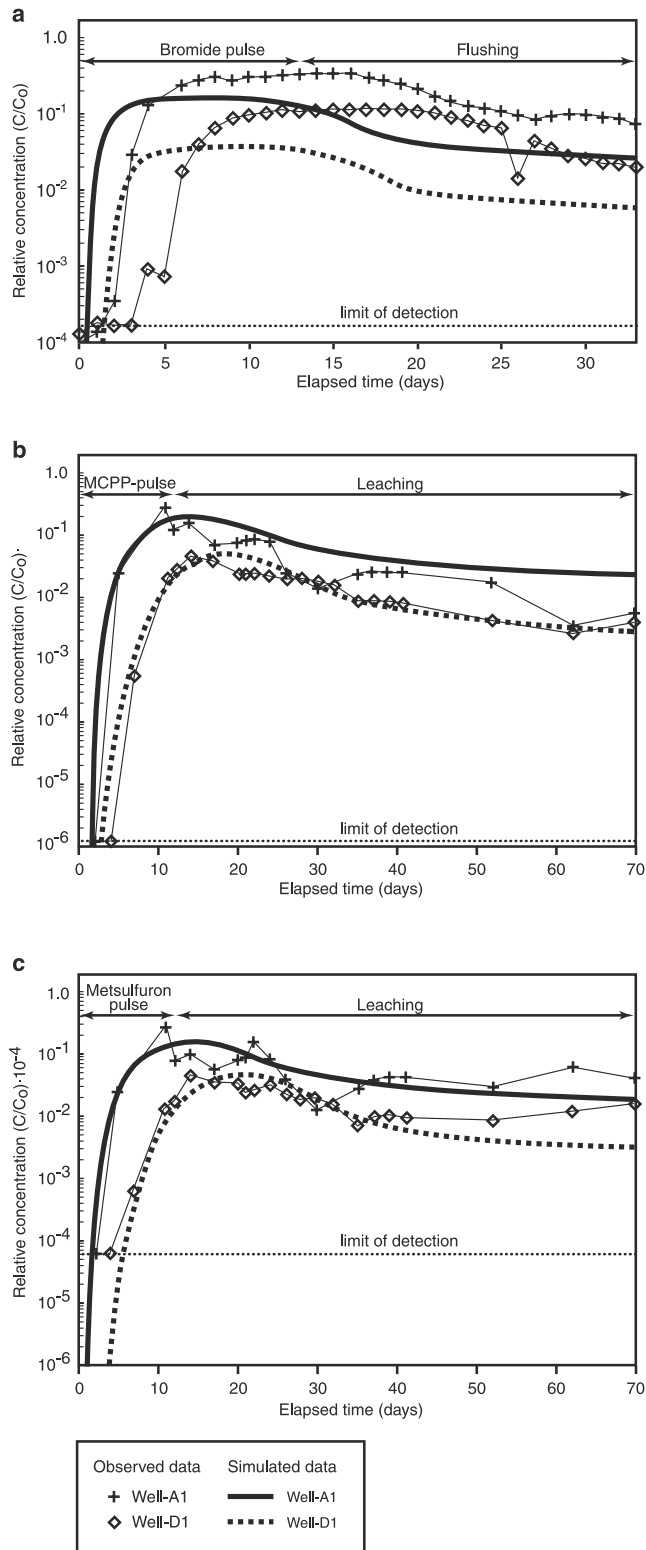


Figure 13. (opposite) Simulated breakthrough of (a) bromide, (b) MCP, and (c) metsulfuron in monitoring wells A1 and D1.

simulated monitoring well concentrations from the two modeling strategies suggest for the specific examples, that after short transport distance along the sand layer the concentrations in monitoring wells are not sensitive to the channels observed in the fractures. Additional simulations

(not shown) of breakthrough from the nondyed fractures, using the LUC aperture values aperture of $13 \mu\text{m}$ (LUC I and II, Table 4), show no breakthrough within the timescale of the experiment, which is in agreement with the observations.

[26] The effective diffusion coefficient (D^*), dispersion in the fractures and matrix, and equilibrium distribution coefficient (K_d) for MCP applied for the till material in the simulations was calibrated using transport data from a fractured LUC. This column was collected from 4–4.5 m depth 50 m north from the current experimental field in a previous study and contained conductive fractures with a mean aperture of $42 \mu\text{m}$ [Jørgensen *et al.*, 1998a]. Using the same hydraulic gradient as applied in the current field experiment this study showed that the DFMD model, calibrated to the flow data, was capable of accurately predicting transport of both chloride and the pesticide MCP at the column (LUC) scale. The same data could not be simulated using a total porosity or effective SP model. The modeling analysis showed that matrix diffusion was sufficient to cause retardation of the nonreactive tracer front ($C/C_0 = 0.5$) approximately 25–80 times compared with the flow rate in the fractures (calculated by the cubic law). Additional retardation of the MCP was consistent with simulations using a retardation factor of 6.8 for sorption of the MCP on the fracture surfaces and the matrix. The role of sorption both on the fracture surfaces and in the matrix was in agreement with concentration profiles measured perpendicular to the fracture surfaces, which showed that the MCP had moved into the matrix, apparently by diffusion [Jørgensen *et al.*, 1998a]. Dispersion parameters for the BSL was calibrated (not shown) from bromide transport data obtained from a separate experiment with this unit [Hoffmann, 2000].

[27] The reasonable approximation by the simulation to the observed transport in the current field experiment suggest that the MCP matrix diffusion, sorption and dispersion values derived from the previous LUC experiment are only to a small extent dependent on the difference in scale between the LUC and the field experiment. This suggests that model calibration using LUC data, which otherwise would be difficult to obtain for the active flow system, may improve model performance prior to field scale simulations.

5. Summary and Conclusions

[28] Water saturated flow and solute transport through the unoxidized zone of a fractured clayey till into a local sandy aquifer was investigated using a natural hydraulic gradient in two experimental fields each 40 m^2 and 2 m deep. The bulk hydraulic conductivity of the two fields was 8×10^{-9} and $15 \times 10^{-9} \text{ m/s}$, respectively, which is normal for clay-rich till aquitards elsewhere in Denmark. Flow in the unfractionated matrix constituted only approximately 2–4% of the total water recharge through the till and represented a flow rate of only a few millimeters per year. It was shown that 96% and 98% of the water recharge into the aquifer occurred along vertical fractures in the till. It was found that fracture channels constituted 10–26% of measured total horizontal fracture lengths. These channel sections conducted 92% of flow in the fractures, while less than 6% occurred along the remaining fractures without channels. The channels were apparently formed by tree roots, which had penetrated along the fractures into the aquifer at 5.5 m depth. From infiltration

of dye tracers the study revealed that the effective fracture spacing (spacing between fracture channel sections) was approximately 5 m, which is 5 times greater than the observed spacing between stained fractures in the till. From the cubic law it was calculated that the mean fracture aperture and flow rates along the fracture channel sections were 49 μm and 100–300 m/d, respectively. It was moreover found from a large undisturbed column (LUC) taken along a single fracture root channel that the aperture of the channel was 124 μm which is corresponding to a flow rate of more than 1 km/d. Corresponding LUC derived values along fracture sections with no channels were 9–22 m/d.

[29] It was found that applied bromide (nonreactive tracer) and pesticides (MCPP and metsulfuron) migrated very rapidly along the fracture root channels into the aquifer with only approximately 50% decline of the applied concentrations. From these results the experiments show that the upper 6 m of the aquitard only provides minimum protection against contaminant transport into the aquifer.

[30] Simulations of flow and transport were carried out with a discrete fracture/matrix diffusion (DFMD) modeling approach for the fractured till coupled with the single porosity (SP) approach for aquifer. It was found that based on the measured effective fracture spacing and mean apertures for conductive fractures in the till, the simulations could reasonably approximate the observed concentrations of bromide and pesticides in the monitoring wells (completed in the aquifer at the margin of the experimental field). It was furthermore found that the applied LUC derived MCPP matrix diffusion, sorption and dispersion values were only to a small extent dependent on the difference in scale between the LUC and the field experiment. This suggests that model calibration using LUC data, which otherwise would be difficult to obtain for the active flow system, may improve model performance prior to field scale simulations.

[31] **Acknowledgments.** This investigation was funded by the Danish Environmental Research Program; "Pesticides and Groundwater" (SMP96) and the EU-research project "Pesticides in European Groundwater: Detailed Study of Representative Aquifers and Simulation of Possible Evolution Scenarios" (PEGASE). The authors would like to thank Danish Geotechnical Institute for using their large undisturbed column (LUC) laboratory.

References

- Broholm, K., P. R. Jørgensen, A. B. Hansen, E. Arvin, and M. Hansen, Transport of creosote compounds in a large, intact clayey till column, *J. Contam. Hydrol.*, 39, 309–329, 1999a.
- Broholm, K., A. B. Hansen, P. R. Jørgensen, E. Arvin, and M. Hansen, Transport and biodegradation of creosote compounds in a large, intact clayey till column, *J. Contam. Hydrol.*, 39, 331–348, 1999b.
- Brown, D. C., R. A. Hodgkinson, D. A. Rose, J. K. Syers, and S. J. Wilcockson, Movement of pesticides to surface waters from a heavy clay soil, *Pesticide Sci.*, 43, 131–140, 1995.
- European Testing Committee (ETC), Standardisation of Laboratory Testing-5, Danish Geotech. Soc., Lyngby, Denmark 1995.
- GEUS, Grundvandsovervågning 1999, Danmarks og Grønlands Geologiske Undersøgelse, Miljø- og Energiministeriet, Copenhagen, 1999.
- Foged, N., and E. Wille, Alteration of clay hydraulic conductivity by various chemical solutions, *Rapp. P7*, Lossepladsprojekt, Copenhagen, 1992.
- Fredericia, J., Saturated hydraulic conductivity of clayey tills and the role of fractures, *Nord. Hydrol.*, 21, 119–132, 1990.
- Helstrup, T., Undersøgelse og matematisk modellering af præferentiel strømning og matrix diffusion i moræner på en skov- og en landbrug-slokalitet, M.Sc. thesis, Geol. Inst., Univ. of Copenhagen, Copenhagen, 1998.
- Helweg, A., and I. Fomsgaard, Nedbrydning og adsorption af phenoxyherbicide (2,4-D, dichlorprop, MCPA og mechlorprop) i og under pløjelaget, in *Phenoxy-syrer: Vurdering af risiko for grundvand*, pp. 17–28, Miljøstyrelsen, Copenhagen, 1996.
- Hinsby, K., L. D. McKay, P. R. Jørgensen, M. Lenczewski, and C. P. Gerba, Fracture aperture measurements and migration of solutes, and immiscible creosote in a column of clay-rich till, *Ground Water*, 34(6), 1065–1075, 1996.
- Hoff, A. G., Radon transport in fractured soil: Laboratory experiments and modelling. Ph.D. thesis, Risø Natl. Lab., Roskilde, Denmark, 1998.
- Hoffmann, M., Feltundersøgelser af hydraulisk ledende makroporer i reduceret moræner – matematisk modellering af konservativ og reaktiv stoftransport, M.Sc. thesis, Geol. Inst., Univ. of Copenhagen, Copenhagen, 2000.
- Jacobsen, P. R., and K. E. Klint, Fracture distribution and occurrence of DNAPL in a clayey lodgement till, *Nord. Hydrol.*, 30(4/5), 285–300, 1999.
- Jensen, M. B., P. R. Jørgensen, H. C. B. Hansen, and N. E. Nielsen, Macropore mediated subsurface transport of dissolved phosphate, *J. Environ. Qual.*, 27, 1130–1137, 1998a.
- Jensen, M. B., H. C. B. Hansen, S. Hansen, P. R. Jørgensen, J. Magid, and N. E. Nielsen, Phosphate solute transport in large monolith of structured, clayey subsoil as affected by ionic strength of infiltrating water, *J. Environ. Qual.*, 27, 39–145, 1998b.
- Jensen, M. B., H. C. B. Hansen, P. R. Jørgensen, and J. Magid, Leaching of phosphate in structured soil—A two step process, *Nord. Hydrol.*, 30(4/5), 361–378, 1999.
- Jørgensen, P. R., Spredning af forurening i moræner, *Miljøstyrelsen, miljøprojekt 155*, Danish Environ. Prot. Agency, Copenhagen, 1990.
- Jørgensen, P. R., and J. Fredericia, Migration of nutrients, pesticides and heavy metals in fractured clayey till, *Geotechnique*, 42, 67–77, 1992.
- Jørgensen, P. R., and N. H. Spliid, Mechanisms and rates of pesticide leaching in shallow clayey till, in *Integrated Soil and Sediment Research: A basis for Proper Protection*, edited by H. J. P. Eijsackers and P. Harms, pp. 247–253, Kluwer Acad., Norwell, Mass., 1993.
- Jørgensen, P. R., and N. H. Spliid, Migration and biodegradation of pesticides in clayey till, *Pesticides Res.* 37, 111 pp., Minist. of Environ. and Energy, Copenhagen, 1998.
- Jørgensen, P. R., L. D. McKay, and N. H. Spliid, Evaluation of chloride and pesticide transport in a fractured clayey till using large undisturbed columns and numerical modeling, *Water Resour. Res.*, 34(4), 539–553, 1998a.
- Jørgensen, P. R., J. Baumann, T. Helstrup, J. Urupog, and K. Butzbach, Hydraulik i sprækket ler, *Geol. Nyt*, 2, 13–16, 1998b.
- Jørgensen, P. R., T. M. Schrøder, G. Felding, A. Helweg, N. H. Spliid, M. Thorsen, J. C. Refsgaard, and O. H. Jacobsen, Validation and development of pesticide leaching models, *Pesticides Res.* 47, 150 pp., Minist. of Environ. and Energy, Copenhagen, 1998c.
- Jørgensen, P. R., M. Hansen, H. Lindgren, and W. Harrar, Point and non-point source leaching of pesticides in a clayey till groundwater catchment, *Pesticides Res.* 52, 150 pp., Minist. of Environ. and Energy, Copenhagen, 2000.
- Klint, K. E. S., and P. Gravesen, Fractures and biopores in Weichselian clayey till Aquitards at Flakkebjerg, Denmark, *Nord. Hydrol.*, 30(4/5), 267–284, 1999.
- Køppen, B., and N. H. Spliid, Determination of acidic herbicides using liquid chromatography with pneumatically assisted electrospray ionization mass spectrometric and tandem mass spectrometric detection, *J. Chromatogr. A*, 803, 157–168, 1998.
- Lide, D. R., *CRC Handbook of Chemistry and Physics: A Ready-Reference Book of Chemical and Physical Data*, 75th ed., CRC Press, Boca Raton, Fla., 1994.
- Lindhard, B., L. Madsen, L. Clausen, I. Fabricius, P. L. Bjerg, M. Broholm, and N. Tuxen, Sorption af pesticider i grundvandsmagasiner, Det Strategiske Miljøforsknings-program—Pesticider og Grundvand, *Miljøforskning* 42, pp. 17–20, Aarhus, Denmark, 2000.
- McKay, L. D., and J. Fredericia, Distribution and origin and hydraulic influence of fractures in a clay rich glacial deposit, *Can. Geotech. J.*, 32, 957–975, 1995.
- McKay, L. D., J. A. Cherry, and R. W. Gillham, Field experiments in a fractured clay till, 1, Hydraulic conductivity and fracture aperture, *Water Resour. Res.*, 29(4), 1149–1162, 1993a.
- McKay, L. D., R. W. Gillham, and J. A. Cherry, Field experiments in a fractured clay till, 2, Solute and colloid transport, *Water Resources Res.*, 29(12), 3879–3890, 1993b.
- McKay, L., J. Fredericia, M. Lenczewski, J. Morthorst, and K. E. S. Klint, Spatial variability of contaminant transport in a fractured till, Avedøre, Denmark, *Nord. Hydrogeol.*, 30(4/5), 333–360, 1999.

- O'Hara, S. K., B. L. Parker, P. R. Jørgensen, and J. A. Cherry, Trichloroethene DNAPL flow and mass distribution in naturally fractured clay, 1, Evidence of fracture aperture variability, *Water Resour. Res.*, 36(1), 135–147, 2000.
- Pankow, J., and J. A. Cherry (Eds.), *Dense Chlorinated Solvents and Other DNAPLs in Groundwater*, Waterloo Press, Waterloo, Ontario, Canada, 1996.
- Snow, D. T., Rock fracture spacings, openings and porosities, *J. Soil Mech. Found. Div. Proc. Am. Soc. Civil Eng.*, 94, 73–91, 1968.
- Spliid, N. H., and B. Køppen, Determination of polar pesticides in groundwater using liquid chromatography-mass spectrometry with atmospheric pressure chemical ionization, *J. Chromatogr. A*, 736, 105–114, 1996.
- Therrien, R., and E. A. Sudicky, Three-dimensional analysis of variably-saturated flow and solute transport in discretely-fractured porous media, *J. Contam Hydrol.*, 23(1–2), 1–44, 1996.
- Thorsen, M., P. R. Jørgensen, G. Felding, O. H. Jacobsen, N. H. Spliid, and J. C. Refsgaard, Evaluation of a stepwise procedure for comparative validation of pesticide leaching models, *J. Environ. Qual.*, 27, 1183–1192, 1998.
- Tomlin, C., (Ed.), *Pesticide Manual: A World Compendium*, Br. Crop Prot. Council, Farnham, UK, 1994.
- Traub-Eberhart, U., K. Henschel, K. Werner, and K. Klein, Influence of different field sites on pesticide movement into sub-surface drains, *Pesticide. Sci.*, 43, 121–129, 1995.
- Worch, E., Eine neue gleichung zur berechnung von diffusionskoeffizienten gelöster stoffe, *Vom Wasser*, 81, 289–297, 1993.
-
- R. Bossi, National Environmental Research Institute, Frederiksborgvej 399, DK-4000 Roskilde, Denmark.
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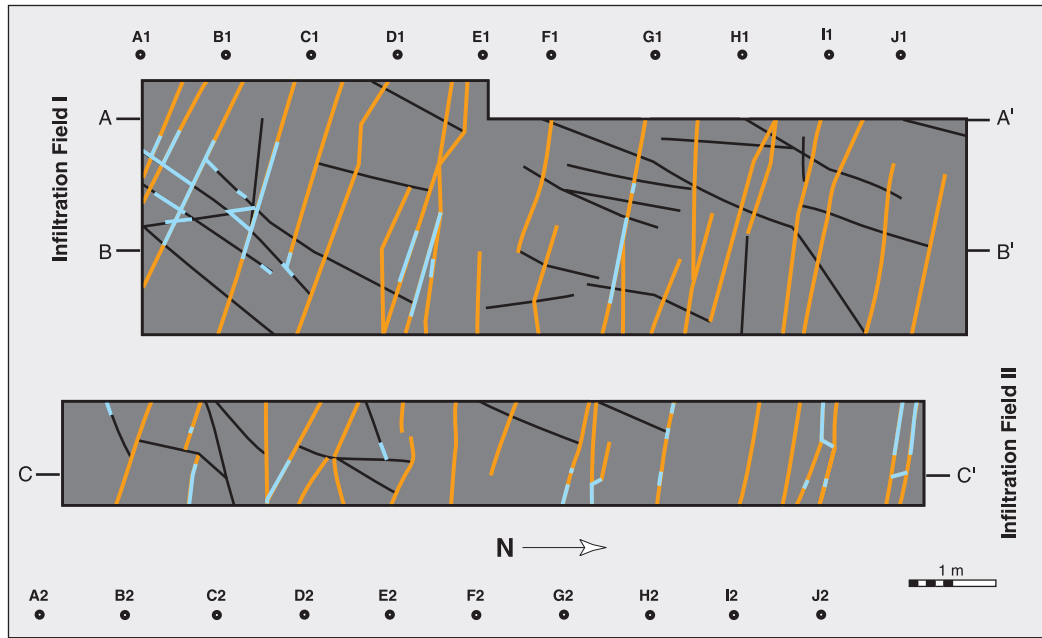
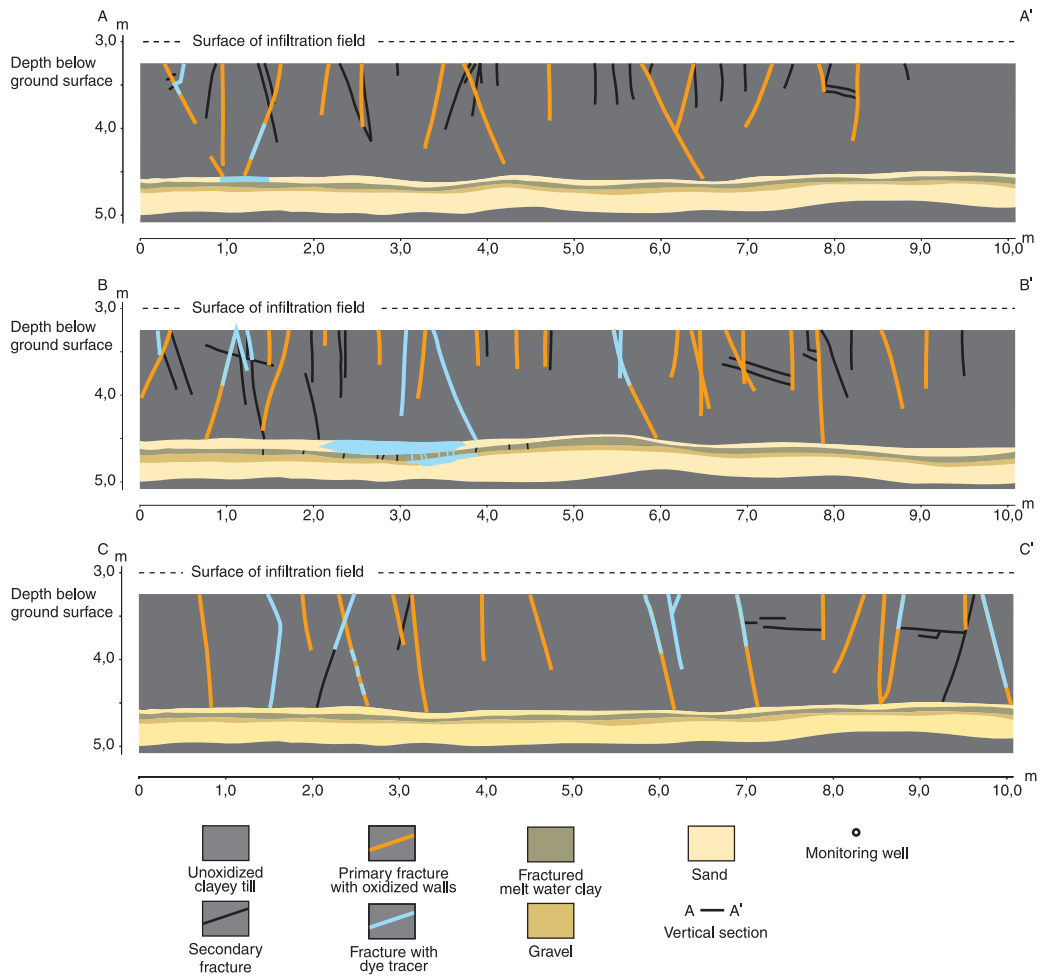
a: Horizontal section**b: Vertical sections**

Figure 8. Distribution of fractures and dyed flow paths in (a) horizontal and (b) vertical sections of experimental fields I and II.

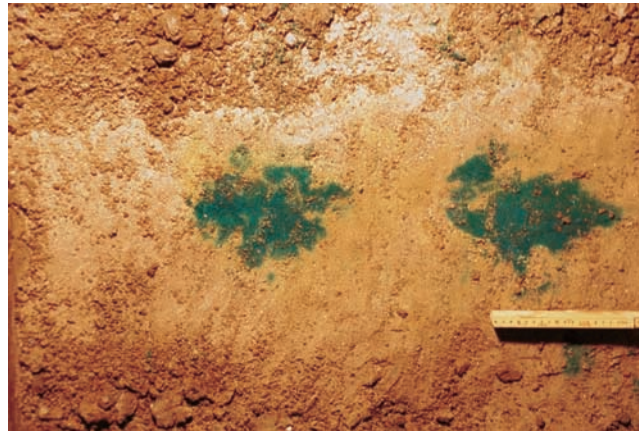


Figure 9. Distribution of dye tracer on exposed horizontal surface representing the boundary between the till and the sandy layer (BSL) in experimental field II. The spot-wise appearance of the dye correlates with fracture root channels observed in the overriding till. Folded yardstick left in the picture is 0.1 m long.

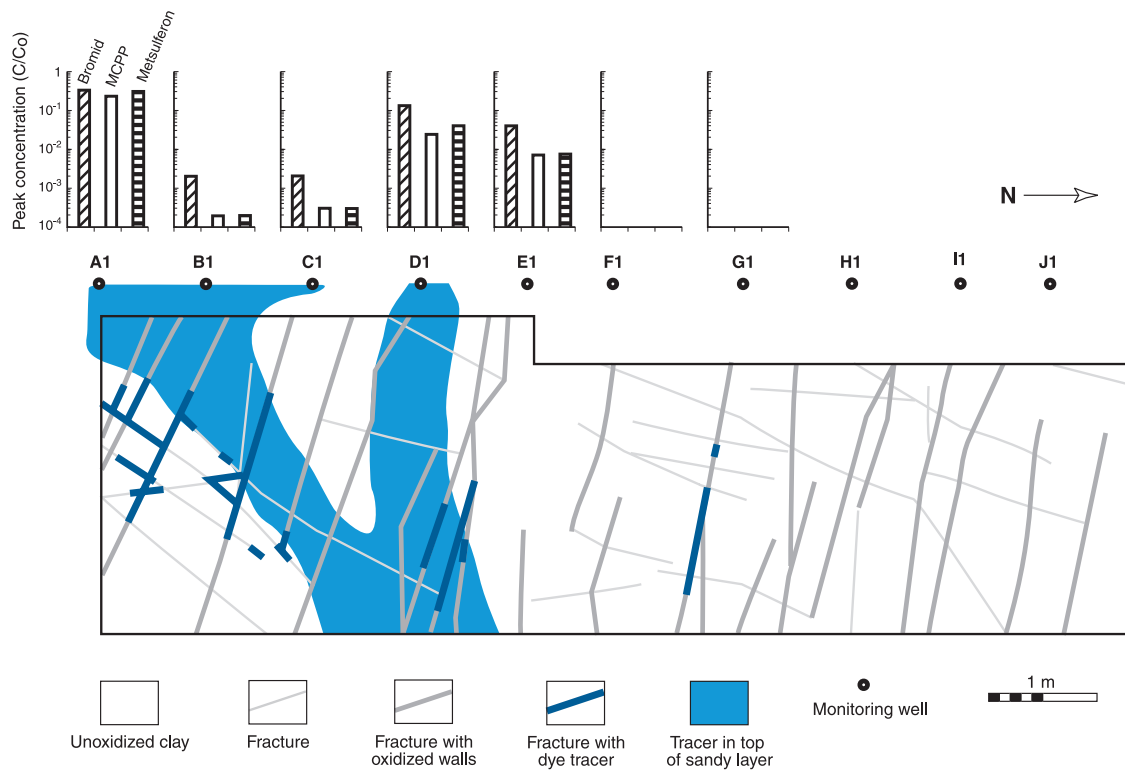


Figure 11. Correlation between dyed flow paths and maximum breakthrough concentration of bromide, MCPP, and metsulfuron in monitoring wells in infiltration field I.